

Lec 23 (practice exercise)

Q(1) Use mathematical induction to prove that $4^n > 3^n + 4$ is true for integral values of $n \geq 2$

Solution:

$$\text{Let } P(n) = 4^n > 3^n + 4$$

Basis step

$$n = 2$$

$$4^2 > 3^2 + 4 = 16 > 13 \quad \text{So } P(2) \text{ is true}$$

Inductive step

Suppose $P(k)$ is true for all $k \geq 2$

$$4^k > 3^k + 4 \quad \longrightarrow (A)$$

We are to prove $P(k+1)$ is true i.e.

$$4^{k+1} > 3^{k+1} + 4$$

$$\text{L.H.S} = 4^{k+1} = 4^k \cdot 4 > 4 \cdot (3^k + 4)$$

$$= 4 \cdot 3^k + 16 \quad \text{by (A) for } k \geq 2$$

$$4^{k+1} > 3^k \cdot (3+1) + 16$$

$$4^{k+1} > 3^k \cdot 3 + 3^k + 16$$

$$4^{k+1} > 3^{k+1} + 3^k + 4 + 12$$

$$4^{k+1} > 3^{k+1} + 4$$

ignoring $12 + 3^k$ as if $a > b + c$

the $a > b$

Hence proved

Question No. 2

Use mathematical Induction to prove that

$$3 + 6 + 9 + \dots + 3n = \frac{3n(n+1)}{2} \text{ for all positive}$$

Integers n .

Solution

$$\text{Let } P(n) = 3 + 6 + 9 + \dots + 3n = \frac{3n(n+1)}{2}$$

Basic Step

$P(n)$ is always equal to 1.

$$\text{So, L.H.S } 3n = 3(1) = 3$$

$$\text{R.H.S } \frac{3n(n+1)}{2} = \frac{3(1)(1+1)}{2} = \frac{6}{2} = 3$$

So $P(1)$ is true.

$$\text{L.H.S} = \text{R.H.S}$$

Inductive Step

Suppose $P(k)$ is true for $k \geq 1$

$$3 + 6 + 9 + \dots + 3k = \frac{3k(k+1)}{2}$$

We replace k at place of n .

To prove $P(k+1)$ true

then replace $(k+1)$ at the place of k .

$$\text{So, } 3 + 6 + 9 + \dots + 3(k+1) = \frac{3(k+1)((k+1)+1)}{2}$$

Prove L.H.S. = R.H.S

$$\text{L.H.S. } 3 + 6 + 9 + \dots + 3k + 3(k+1)$$

$$\left\{ \begin{array}{l} 3 + 6 + 9 + \dots + 3k = \frac{3k(k+1)}{2} \\ \text{In Question} \\ \text{So put this value,} \end{array} \right.$$



$$= 3 + 6 + 9 + \dots + 3k + 3(k+1)$$

$$= \frac{3(k+1)}{2} + 3(k+1)$$

take Common $3(k+1)$,

$$3(k+1) \left(\frac{k}{2} + 1 \right)$$

$$3(k+1) \left(\frac{k+2}{2} \right)$$

$$\frac{3(k+1)(k+2)}{2}$$

$$\frac{3(k+1)(k+1+1)}{2} = \text{R.H.S}$$

So, we have proved L.H.S = R.H.S

Question No: 3

Suppose $n^3 - n$ is (positive) divisible by 6 is true for any positive Integers. Justify the statement for $n \in \mathbb{N}$ $n = k+1$

Solution

Inductive Step

$p(n)$ replace k .

$$n = k$$

$$n^3 - n = k^3 - k$$

$k^3 - k$ is divisible by 6.

Mean that

$$k^3 - k = 6q$$

We need to prove $P(k+1)$ So,
replace $(k+1)$ at k .

$$k^3 - k = 6q \quad (1)$$

$$(k+1)^3 - (k+1) = 6$$

$$L.H.S = (k+1)^3 - (k+1)$$

$$R.H.S = 6q$$

Prove L.H.S is equal to R.H.S

So, L.H.S

$$(k+1)^3 - (k+1)$$

$$(k+1)^3 = k^3 + 3k^2 + 3k + 1$$

$$k^3 + 3k^2 + 3k + 1 - (k+1)$$

$$k^3 + 3k^2 + 3k + 1 - k - 1$$

$$k^3 - k + 3k(k+1)$$

$$6q + 3k(k+1) \quad (2)$$

Now k is an Integer. So

$k(k+1)$ being product of two consecutive integers is an even number

$$\text{Let as } k(k+1) = 2r$$

Now equation 2 can be written as,

$$= 6q + 3 \cdot 2r$$

$$= 6q + 6r$$

$$6(q+r)$$

$$q, r \in \mathbb{Z}$$

$(k+1)^3 - (k+1)$ is divisible by 6.

~~Q.E.D.~~

Now we see this
 $k^3 - k$ is equal to
 $6q$, So put $6q$
 in place of
 $k^3 - k$

③

Question No. 4

Suppose $5^n - 1$ is divisible by 4 is true for $n = k$, Justify $n = k+1$.

Solution

Basic step always true for $n=1$

$$5^1 - 1 = 4$$

$$5^1 - 1 = 4$$

$$4 = 4,$$

Inductive step.

$$n = k$$

$$5^n - 1 = 4$$

$$5^k - 1 = 4 \quad \text{--- (1)}$$

We need to prove $P(k+1)$.

So, $5^{k+1} - 1 = 4q \quad \text{--- (2)}$

$$5^{k+1} - 1$$

$$= 5 \cdot 5^k - 1$$

proved it = $5 \cdot 5^k - 1$

$$= 5^k \cdot (4+1) - 1$$

$$= 5^k \cdot 4 + 5^k - 1 \quad \text{--- (2)} \quad \boxed{5^k - 1 = 4q}$$

$$= 5^k \cdot 4 + 5^k - 1 = 5^k \cdot 4 + 4q$$

$$= 4(5^k + q)$$

Hence $5^{k+1} - 1$ is divisible by 4. Q.E.D.

⑥ Question No: 5

proved that if x divide by y and y divide by z . then prove x divide by z ?

Solution.

By divisibility Definition.

k_1 and k_2 are natural number.

So

$$y = x \rightarrow x \text{ divided by } y.$$

$$z = y \rightarrow y \text{ divide by } z.$$

In this way

$$y = xk_1$$

$$z = yk_2$$

then,

$$z = yk_2, \quad \boxed{y = xk_1} \text{ put it}$$

$$z = xk_1k_2$$

Let k_1 and k_2 is k , so,

Bcz k is natural number.

$$z = xk,$$

$$\cancel{z = k} \boxed{z = x} \rightarrow x \text{ divided by } z,$$

$$z = xk, \quad k \text{ is natural number}$$

So proven that $z = xk$.

Question No: 6

Suppose that if a and b is even in plus mean $(a+b)$ is even, then prove that $(a-b)$ is also even.

Solution

$$\text{take } a+b = 2k$$

$a+b$ two even Integers

$$a = 2k - b$$

Subtract b from both side,

$$a - b = 2k - b - b$$

$$a - b = 2k - 2b$$

$$a - b = 2(k - b)$$

$a - b$ is also even

~~not~~ Hence Proved

Practice Question For lec 26-28
14th 202-

Farha Shah

Question .1

proved by contradiction that

$4 + 3\sqrt{2}$ is an Irrational number.

Solution

In Question we have to prove $4 + 3\sqrt{2}$ is Irrational. So, we let $4 + 3\sqrt{2}$ is rational number.

Let

$4 + 3\sqrt{2}$ is rational number.

So by definition of Rational number
It can be written as ,

$$4 + 3\sqrt{2} = \frac{a}{b}$$

$$4 + 3\sqrt{2} = \frac{a}{b}$$

$$3\sqrt{2} = \frac{a}{b} - 4$$

$$\sqrt{2} = \frac{a - 4b}{3b} = \frac{a - 4b}{3b} \neq 0$$

We see that;

$\sqrt{2}$ is rational number. But if ans $\frac{a - 4b}{3b}$ the value of $\sqrt{2}$ is irrational number by contradiction.

So, $4 + 3\sqrt{2}$ is incorrect in term of Contradiction at irrational number.

$4 + 3\sqrt{2}$ is rational number $\checkmark$

Question No: 2

Use contradiction to prove that if $5n+1$ is odd then n is even.

Solution

We want to prove by contradiction that if $5n+1$ is odd then n is even. Suppose for some integers $5n+1$ is odd but n is not even.

If n is not even it will be odd.

$n = 2m+1$ for some integers

$$5n+1 = 5(2m+1)+1$$

$$10m+5+1$$

$$10m+6$$

$$2(5m+3)$$

Question No: 3

Use contrapositive to prove that if n^2 is odd then n is odd.

Solution

By contraposition if n^2 is odd then n is odd, then we prove that if n is even then n^2 is even.

Let n is even, multiple of 2

$$n = 2k$$

$$n^2 = 2k^2, 2(2k^2)$$

n is multiple of ~~2~~ 2 (n is odd)

Question 4.

Write pre and post Condition of algorithm to find quotient and remainder of division of one positive integers by another.

Solution

Pre-Condition

The input variable a and b are positive integers.

Post-Condition.

The output variable q and r positive integers such that $a = bq + r$ and $0 \leq r < b$.

Question 5.

Use Euclidean algorithm to find $\text{GCD}(6114, 94)$.

Solution. We are to find $\text{GCD}(6114, 94)$.

Divide ~~94~~ 6114 by 94. $94 \overline{) 6114}$

6114	
564	
474	
470	
4	= Remainder = 14,

then Divide 94 by 14 = 10

then Divide 14 by 10 = 2

then Divide 10 by 2 = 0,

So remainder is 2.

Practice Question Lec 29-31

14th 202

Farha Shah

Question 1

There are 3 ^{uses} ~~ways~~ that travel from A to city B and 5 buses travel from city B to city C. Find the number of ways a person can travel a round trip from city A to city C by way of city B.

Solution

For round trip

the person will travel from city A to B, B to C and B to A.

$$A \rightarrow B \rightarrow C \rightarrow B \rightarrow A$$

3 person Buses travel from A to B.
and 5 buses travel from B to C.

So,
$$\begin{array}{ccccccc} A & \rightarrow & B & \rightarrow & C & \rightarrow & B \rightarrow A \\ | & & | & & | & & | \\ 3 & & 5 & & 5 & & 3 \end{array}$$

$$3 \times 5 \times 5 \times 3 = 225$$

way round trip.

Question 2

A bit String is sequence of 0's and 1's. How many bit strings are there of length 7?

Solution

Each bit strings (binary digit) is either 0 or 1.

So there are two ways to choose, each ways.

For bit strings of length seven bits, use product rule.

$$2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 2^7 = 128$$

there are 128 ways length 7.

Question 3

Find n if $P(n, 2) = 42$.

$$P(n, 2) = 42,$$

$$\frac{n!}{(n-2)!} = 42$$

$$\frac{n(n-1)(\cancel{n-2}!) }{(n-2)!} = 42$$

$$n(n-1) = 42$$

$$n^2 - n = 42$$

$$n^2 - n - 42,$$

By factorization,

$$n^2 - 7n + 6n - 42,$$

$$n(n-7) + 6(n-7)$$

$$(n-7)(n+6)$$

$$n-7=0, \quad n+6=0,$$

$$n=7, \quad n=-6$$

Ignore negative value, value of n is 7

44

Question : 4

There are 10 boys and 15 ~~boy~~ girls in class. In how many ways can a quiz team of 5 boys and 5 girls be formed from this class?

Solution.

5 boys can be selected from 10 in following number of ways of Boys,

$${}^{10}C_5 = 252,$$

And

5 girls can be selected from 15 in following number of ways of Girls,

$${}^{15}C_5 = 3003,$$

$$\begin{array}{r} {}^{10}C_5 \\ \hline \frac{10!}{5!(10-5)!} \\ \hline \frac{10!}{5!} \\ \hline \frac{10 \times 9 \times 8 \times 7 \times 6 \times 5!}{5! \times 5!} \\ \hline \frac{10 \times 9 \times 8 \times 7 \times 6}{5 \times 4 \times 3 \times 2 \times 1} \\ \hline 2 \times 3 \times 2 \times 3 \times 7 = 252 \end{array}$$

$$\begin{aligned} \text{Total ways} &= {}^{10}C_5 \times {}^{15}C_5 \\ &= 252 \times 3003 \\ &= 756756, \end{aligned}$$